ASSIGNMENT-BANKING SYSTEM
TASK 4
Subquery and its type

1. Retrieve the customer(s) with the highest account balance.

```
Database changed
mysql> SELECT C.customer_id, C.first_name, C.last_name, A.account_id, A.account_type, A.balance
    -> FROM Customers C
    -> JOIN Accounts A ON C.customer_id = A.customer_id
    -> WHERE A.balance = (SELECT MAX(balance) FROM Accounts);
+-------------+------------+-----------+------------+--------------+-----------+
| customer_id | first_name | last_name | account_id | account_type | balance   |
+-------------+------------+-----------+------------+--------------+-----------+
|           4 | Sirius     | Black     |          4 | current      | 777700.00 |
+-------------+------------+-----------+------------+--------------+-----------+
1 row in set (0.15 sec)
```

2. Calculate the average account balance for customers who have more than one account.

```
mysql> SELECT AVG(A.balance) AS avg_balance
    -> FROM Accounts A
    -> WHERE A.customer_id IN (
    ->     SELECT customer_id
    ->     FROM Accounts
    ->     GROUP BY customer_id
    ->     HAVING COUNT(account_id) > 1
    -> );
+-------------+
| avg_balance |
+-------------+
| 6533.458333 |
+-------------+
1 row in set (0.04 sec)
```

3. Retrieve accounts with transactions whose amounts exceed the average transaction amount.

```
mysql> SELECT T.*, A.account_type
    -> FROM Transactions T
    -> JOIN Accounts A ON T.account_id = A.account_id
    -> WHERE T.amount > (SELECT AVG(amount) FROM Transactions);
+----------------+------------+------------------+---------+---------------------+--------------+
| transaction_id | account_id | transaction_type | amount  | transaction_date    | account_type |
+----------------+------------+------------------+---------+---------------------+--------------+
|              4 |          4 | transfer         | 1500.00 | 2024-03-23 11:10:00 | current      |
|              6 |          6 | deposit          | 2500.00 | 2024-03-25 12:00:00 | current      |
|              8 |          8 | transfer         | 1800.00 | 2024-03-27 15:05:00 | current      |
|              9 |          9 | deposit          | 5000.00 | 2024-03-28 08:30:00 | savings      |
|             11 |          1 | deposit          | 2000.00 | 2024-03-30 10:30:00 | savings      |
|             13 |         12 | deposit          | 3000.00 | 2024-03-29 14:15:00 | savings      |
|             14 |         13 | transfer         | 1500.00 | 2024-03-28 09:00:00 | current      |
+----------------+------------+------------------+---------+---------------------+--------------+
7 rows in set (0.03 sec)
```

4. Identify customers who have no recorded transactions.

```
mysql> SELECT C.customer_id, C.first_name, C.last_name
    -> FROM Customers C
    -> WHERE C.customer_id NOT IN (
    ->     SELECT DISTINCT A.customer_id
    ->     FROM Accounts A
    ->     JOIN Transactions T ON A.account_id = T.account_id
    -> );
+-------------+------------+-----------+
| customer_id | first_name | last_name |
+-------------+------------+-----------+
|          11 | Draco      | Malfoy    |
+-------------+------------+-----------+
1 row in set (0.00 sec)
```

5. Calculate the total balance of accounts with no recorded transactions.

```
mysql> SELECT SUM(A.balance) AS total_balance_without_transactions
    -> FROM Accounts A
    -> WHERE A.account_id NOT IN (
    ->     SELECT DISTINCT T.account_id FROM Transactions T
    -> );
+-------------------------------------+
| total_balance_without_transactions |
+-------------------------------------+
|                             2500.00 |
+-------------------------------------+
1 row in set (0.01 sec)
```

6. Retrieve transactions for accounts with the lowest balance.

```
mysql> SELECT T.*
    -> FROM Transactions T
    -> WHERE T.account_id IN (
    ->     SELECT account_id FROM Accounts WHERE balance = (SELECT MIN(balance) FROM Accounts)
    -> );
+----------------+------------+------------------+--------+---------------------+
| transaction_id | account_id | transaction_type | amount | transaction_date    |
+----------------+------------+------------------+--------+---------------------+
|             10 |         10 | withdrawal       | 300.00 | 2024-03-29 19:55:00 |
+----------------+------------+------------------+--------+---------------------+
1 row in set (0.01 sec)
```

7. Identify customers who have accounts of multiple types.

```
mysql> SELECT C.customer_id, C.first_name, C.last_name, COUNT(DISTINCT A.account_type) AS account_types
    -> FROM Customers C
    -> JOIN Accounts A ON C.customer_id = A.customer_id
    -> GROUP BY C.customer_id
    -> HAVING COUNT(DISTINCT A.account_type) > 1;
+-------------+------------+-----------+---------------+
| customer_id | first_name | last_name | account_types |
+-------------+------------+-----------+---------------+
|           1 | Harry      | Potter    |             2 |
|           5 | Remus      | Lupin     |             2 |
+-------------+------------+-----------+---------------+
2 rows in set (0.03 sec)
```

8. Calculate the percentage of each account type out of the total number of accounts.

```
mysql> SELECT account_type,
    ->        COUNT(*) AS total_accounts,
    ->        (COUNT(*) * 100.0 / (SELECT COUNT(*) FROM Accounts)) AS percentage
    -> FROM Accounts
    -> GROUP BY account_type;
+--------------+----------------+------------+
| account_type | total_accounts | percentage |
+--------------+----------------+------------+
| savings      |              7 |   50.00000 |
| current      |              6 |   42.85714 |
| zero_balance |              1 |    7.14286 |
+--------------+----------------+------------+
3 rows in set (0.01 sec)
```

9. Retrieve all transactions for a customer with a given customer_id.

```
mysql> SELECT T.*
    -> FROM Transactions T
    -> JOIN Accounts A ON T.account_id = A.account_id
    -> WHERE A.customer_id = 1;
+----------------+------------+------------------+---------+---------------------+
| transaction_id | account_id | transaction_type | amount  | transaction_date    |
+----------------+------------+------------------+---------+---------------------+
|              1 |          1 | deposit          |  500.00 | 2024-03-20 10:15:00 |
|             11 |          1 | deposit          | 2000.00 | 2024-03-30 10:30:00 |
|             12 |          1 | withdrawal       |  500.00 | 2024-03-31 12:45:00 |
|             15 |         11 | withdrawal       | 1000.00 | 2024-03-27 16:30:00 |
+----------------+------------+------------------+---------+---------------------+
4 rows in set (0.01 sec)
```

10. Calculate the total balance for each account type, including a subquery within the SELECT clause.

```
mysql> SELECT account_type,
    ->        (SELECT SUM(balance) FROM Accounts A2 WHERE A2.account_type = A1.account_type) AS total_balance
    -> FROM Accounts A1
    -> GROUP BY account_type;
+--------------+---------------+
| account_type | total_balance |
+--------------+---------------+
| savings      |      35900.75 |
| current      |     811200.75 |
| zero_balance |          0.00 |
+--------------+---------------+
3 rows in set (0.01 sec)
```

NOTES:

Additionally a record is inserted in both the Customers and Accounts table.

Customers table:

```
mysql> select * from Customers;
+-------------+------------+-----------+-----------+---------------------------+--------------+---------------------------------------+
| customer_id | first_name | last_name | DOB       | email                     | phone_number | address                               |
+-------------+------------+-----------+-----------+---------------------------+--------------+---------------------------------------+
|           1 | Harry      | Potter    | 1990-07-31 | harry.potter@gmail.com    | 9876543210   | 4 Privet Drive, Little Whinging, Surrey |
|           2 | Ron        | Weasley   | 1990-03-01 | ron.weasley@gmail.com     | 9845678901   | The Burrow, Ottery St. Catchpole, Devon |
|           3 | Hermione   | Granger   | 1989-09-19 | hermione.granger@gmail.com | 9786543210   | Hampstead Garden Suburb, London       |
|           4 | Sirius     | Black     | 1959-11-03 | sirius.black@gmail.com    | 9765432109   | 12 Grimmauld Place, London            |
|           5 | Remus      | Lupin     | 1960-03-10 | remus.lupin@gmail.com     | 9876504321   | 13th Avenue Road,London               |
|           6 | Lily       | Evans     | 1960-01-30 | lily.evans@gmail.com      | 9876543098   | Spinner?s End, Cokeworth              |
|           7 | Luna       | Lovegood  | 1981-02-13 | luna.lovegood@gmail.com   | 9785463210   | Lovegood House, Ottery St. Catchpole  |
|           8 | Narcissa   | Malfoy    | 1955-06-01 | narcissa.malfoy@gmail.com | 9898765432   | Malfoy Manor, Wiltshire               |
|           9 | James      | Potter    | 1960-03-27 | james.potter@gmail.com    | 9876234510   | Godric?s Hollow, West Country, England |
|          10 | Tom        | Riddle    | 1926-12-31 | tom.riddle@gmail.com      | 9887766543   | Wool?s Orphanage, London              |
|          11 | Draco      | Malfoy    | 1980-06-05 | draco.malfoy@gmail.com    | 9876543111   | Malfoy Manor, Wiltshire               |
+-------------+------------+-----------+-----------+---------------------------+--------------+---------------------------------------+
11 rows in set (0.00 sec)
```

Accounts table:

account_id	customer_id	account_type	balance
1	1	savings	2000.00
2	2	current	3200.50
3	3	savings	7800.75
4	4	current	777700.00
5	5	savings	5400.00
6	6	current	9000.25
7	7	savings	1200.00
8	8	current	4300.00
9	9	savings	10000.00
10	10	zero_balance	0.00
11	1	current	5000.00
12	3	savings	7000.00
13	5	current	12000.00